

Aditi Saxena

Education

Indian Institute of Science, Bengaluru	Postgraduate	Aug 2024 - Present
- Ph.D. in Brain, Computation and Data Science - Coursework: Stochastic Models, Matrix Theory, Digital Signal Processing, Machine Learning, Optimization for ML, Statistics		8.30/10
Indian Institute of Technology Goa	Undergraduate	Aug 2018 - May 2022
- Bachelor of Technology in Computer Science and Engineering - Undergraduate Thesis: Multi Parameter Learning Multi-Armed Bandit - Undergraduate Coursework: Artificial Intelligence, Machine Learning, Data Analysis, Operating Systems, Computer Architecture, Optimization, Data Structures, Algorithms, Discrete Structures, Automata Theory, Computer Networks - Teaching Assistant: English Language Proficiency Program		9.29/10 9.67/10
City Montessori School, Gomtinagar, Lucknow	Grade 12	98% (All India Rank 08)
City Montessori School, Aliganj, Lucknow	Grade 10	97% (All India Rank 12)

Employement

Microsoft IDC, Surface ASPX, Hyderabad	Software Engineer	Aug 2022 - Jul 2024
- Pursued and completed five impactful epics with distinct technological frameworks and codebases - Designed and deployed an internal web-portal for automating viewing and editing of nested configuration data (<i>Blazor Server</i>) - Incorporated OAuth2 client credentials authentication flow to ensure secure data access (<i>Azure AD, Managed Identities, MSAL</i>) - Performed the Microsoft Security Development Lifecycle for the web portal (<i>Threat Modelling, Privacy Review, Accessibility</i>) - Contributed to the development of an internal web portal for tracking team tasks and facilitating sprint planning (<i>Blazor, ASP.NET,JS</i>) - Worked across teams to add OS update speedbumps, to decrease users contacting support (<i>React, TypeScript, WebView2</i>) - Created a microservice for retrieving and updating subsystem registry values through RPC calls using Native Operability in C# - Deduced and addressed inefficiencies in asset fetch calls by transitioning caching from LiteDB to App Storage (<i>Windows.Storage</i>) Hackathon Projects - Interviewer Copilot, Customer Voice, Colleague Castle (<i>WinAppSDK, WinUI3, Azure Open AI, Blazor</i>)		
Otsuka Corporation, Tokyo, Japan	AI Developer Intern	Jun 2021 - Jul 2021
- Implemented Deep Learning object detection models YOLO, DETR and segmentation model SOLO (<i>PyTorch, TensorFlow</i>) - Assessed the accuracy, inference speed and visualization capability of each model on the Microsoft-COCO dataset - Optimized DETR by fine-tuning it on OCHuman dataset containing 5081 images to detect human objects with 68% accuracy - Designed and developed web application interface to deploy the optimized model (<i>HTML, CSS, Javascript, Flask, TensorFlow.js</i>)		

Publication

Saxena, Aditi, and Sharad Sinha. "Machine Learning Based Webcasting Analytics for Indian Elections-Reflections on Deployment." *International Conference on Computer Vision and Image Processing*. Cham: Springer Nature Switzerland, 2022.

Graduate Research Experience

Predict Brain Encoding from Multi-modal Stimuli	Jan 2025 - Present
- Collaborated in a team of two for the Algonauts Challenge 2025, focused on modeling brain responses to multimodal stimuli (text, audio, video) using the CNeuromod fMRI dataset (<i>Python</i>) - Designed and implemented a multimodal feature extraction pipeline using pretrained models: BERT for textual input, Wav2Vec2 for audio, and SlowFast networks for video to obtain temporally-aligned representations (<i>PyTorch</i>) - Developed a representation learning framework combining Contrastive Predictive Coding (CPC) with LSTM encoders, trained to predict brain responses with correlation loss, achieving baseline-level performance - Conducted exploratory research on Hemodynamic Response Function (HRF) modeling, employing Deep Variational Bayes filters and Gamma Function parameterizations to account for individual and regional variation in fMRI response latency and shape	

EEG to Visual Stimuli Latent Space

Aug 2024 - Dec 2024

- Developed a deep learning pipeline using ResNet-18 architecture to extract features from multi-electrode EEG spectrograms
- Trained the model on a large-scale EEG dataset of visual object recognition to achieve upto 62% correlated reconstruction of visual stimuli feature vectors, across multiple subjects (*Python, TensorFlow*)
- Incorporated subject and trial embeddings into the architecture to promote generalization across participants and sessions, improving robustness and transferability

Implementation and Benchmarking of FL Algorithms Course Project

Apr 2025

- Conducted a comparative study of FedAvg and SCAFFOLD algorithms for federated learning using CIFAR-10 and Fashion-MNIST datasets, focusing on client drift and data heterogeneity
- Implemented and evaluated models under varying hyperparameters; found SCAFFOLD offers comparable accuracy to FedAvg but is more sensitive to learning rate and local update frequency (*Python, PyTorch*)

Graph Fourier Transform

Course Project

Oct 2024 - Nov 2024

- Implemented Graph Fourier Transform (GFT) to extract spectral features from EEG data for classifying Alzheimer's, frontotemporal dementia, and healthy subjects; achieved up to 82% accuracy improvement using GFT-enhanced features (*MATLAB*)
- Compared performance of SVM, KNN, Random Forest, Neural Networks with and without GFT features, demonstrating the effectiveness of graph-based signal representations (*Statistics and Machine Learning Toolbox*)
- Explored preprocessing pipelines (ICA, ASR, bandpass filtering) and application of Graph Signal Processing in brain data analysis (*Signal Processing Toolbox*)

Undergraduate Technical Experience

Multi-Parameter Learning Multi-Armed Bandit

Undergraduate Thesis

Sep 2021 - Apr 2022

- Explored classic stochastic multi-armed bandit framework for sequential decision making with regret analysis
- Implemented existing allocation algorithms UCB1, ϵ -greedy, Thompson Sampling - Bernoulli and Gaussian (*Python3, Numpy*)
- Verified the implementation by corroborating results with publicly available data for 10 arms and variable time steps
- Analyzed different variations namely Choice bandits, Ballooning bandits, Recovering Bandits, Mortal bandits
- Conducted the regret upper and lower bound analysis for Bi-Parameter learning for Online Crowdsourcing theoretically
- Worked on extending the model and regret analysis to multi-parameter learning multi-armed bandit

Election Monitoring System

Research Project

Nov 2021 - Mar 2022

- Collaborated with Goa Election Commission to develop and deploy a polling monitoring system for the Assembly Elections-2022
- Implemented Deep Learning models YOLOv5 and DeepSORT for human object detection and tracking (*Python3, PyTorch*)
- Created a real-time video capturing and publishing server using the Real Time Streaming Protocol (*OpenCV, RTSP, FFmpeg*)
- Optimized the system to process 200 live videos simultaneously and transmit back the output data (*CUDA, REST API*)

Improved Protein Structure Prediction

Research Project

Jun 2020 - Jan 2021

- Tackled Protein Structure Prediction problem by combining ab-initio Machine Learning and template-based methods
- Performed sequence alignment, secondary structure prediction with opensource tools (*PyMol, Clustal Omega, BLAST, HHpred*)
- Explored the state-of-the-art protein structure prediction methods like DeepMind's AlphaFold and ProSPr
- Contributed to the development of library for Protein Threading by Template-based approach (*Python3, Z3, Numpy*)

Additional Experience

- **Leadership:** Google Developer Student Club Lead, Editor of college newsletter - Gurukul Varta, Internship Coordinator
- **Achievements:** Outstanding grade (AP) in Artificial Intelligence, Computer Architecture, Computer Programming courses, Selected for IBM Research's Maitreyee 2021, Google Solve.Code 2021, Sakura Science Student Exchange Program 2017
- **MOOCs:** Python, Intro to Machine Learning, Neural Networks, Computer Vision, Cuda Python, Deep Learning, Google Cloud

Software skills

- Languages: C#, Python, Java, C++, SQL, R, Octave, Latex, Bash, HTML, CSS, MIT/GNU Scheme, VHDL
- Libraries: MSAL, Microsoft Graph API, NumPy, pandas, Matplotlib, Keras, TensorFlow, PyTorch, OpenGL,